PHISHING EMAIL
INVESTIGATION REPORT

Analyst: Amit Pal
Report ID: PHISH-2026-001
Date: March 2026
Source: PhishingPot Dataset (GitHub)

RISK LEVEL: HIGH — CONFIRMED PHISHING

Tools Used: CyberChef | AbuseIPDB | WHOIS/ARIN | MXToolbox | VirusTotal | URLScan.io

1. Executive Summary

A sophisticated phishing email was detected impersonating Banco Bradesco's Livelo Rewards Program targeting Portuguese-speaking banking customers. The attack uses social engineering — creating urgency around expiring reward points — to trick victims into clicking a malicious link that leads to a credential-harvesting landing page.

This report documents the complete analysis of the phishing email including header analysis, IP investigation, authentication failures, URL analysis, and indicators of compromise (IOCs). All findings are mapped to the MITRE ATT&CK framework.

Field	Value
Report ID	PHISH-2026-001
Analyst	Amit Pal
Date Analysed	March 2026
Email Source	PhishingPot Dataset — github.com/rf-peixoto/phishing_pot
Attack Target	Portuguese-speaking banking customers (Banco Bradesco)
Attack Goal	Credential harvesting — redirect to phishing landing page
Risk Level	HIGH — Confirmed Phishing
MITRE Techniques	T1566.001, T1598.003, T1204.001, T1056.003

2. Email Details & Header Analysis

2.1 Basic Email Information

Field	Value
Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
From (Display)	BANCO DO BRADESCO LIVELO <banco.bradesco@atendimento.com.br>
Return-Path	root@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
To	phishing@pot (PhishingPot honeypot)
Date	Tuesday, 19 September 2023 18:35:49 UTC
Content-Type	text/html; charset=UTF-8
Encoding	Base64 — obfuscated to evade detection

2.2 Finding #1 — Sender Domain Spoofing

CRITICAL: The sender claims to be Banco Bradesco but uses the domain atendimento.com.br — NOT the legitimate bradesco.com.br domain. This is classic domain spoofing.

Analysis Point	Finding
Claimed identity	BANCO DO BRADESCO LIVELO (legitimate Brazilian bank)
Actual sender domain	atendimento.com.br — NOT the bank's official domain
Legitimate bank domain	bradesco.com.br — completely different
Return-Path hostname	ubuntu server — cloud VPS, not a bank mail server
Verdict	SPOOFED — attacker registered a lookalike domain to deceive victims

2.3 Finding #2 — Email Authentication Failures

ALL 4 email authentication checks FAILED — this email has NO legitimate origin. A real bank email would pass SPF, DKIM and DMARC.

Authentication Check	Result

SPF (Sender Policy Framework)	FAILED — temperror — sending server not authorised
DKIM (DomainKeys Identified Mail)	NONE — email not digitally signed — easily forged
DMARC (Domain-based Auth)	FAILED — temperror — domain policy violated
CompAuth (Composite Auth)	FAILED — Microsoft composite check failed
Overall Verdict	HIGHLY SUSPICIOUS — No authentication whatsoever

Explanation: SPF checks whether the sending server is authorised to send on behalf of the domain. DKIM provides a cryptographic signature proving the email was not tampered with. DMARC ties both together and defines policy for failures. All three failing together means this email has absolutely no legitimate origin.

3. IP Address & Infrastructure Analysis

3.1 Sender IP Address

X-Sender-IP: 137.184.34.4

Field	WHOIS Result
IP Address	137.184.34.4
IP Range	137.184.0.0 — 137.184.255.255
Organisation	DigitalOcean, LLC
ASN	AS14061
Location	Santa Clara, California, USA
Usage Type	Data Center / Web Hosting / Transit
Registration Date	2019-11-13
Verdict	Cloud VPS server — NOT a legitimate bank mail server

3.2 Finding #3 — Suspicious Cloud Infrastructure

A real bank like Banco Bradesco would send emails from their own mail servers (mail.bradesco.com.br etc). This email originates from a DigitalOcean rented VPS — a classic attacker tactic to quickly spin up and discard phishing infrastructure.

3.3 AbuseIPDB Reputation Check

Field	Result
IP Address	137.184.34.4
Reports Found	10 reports in database
Abuse Confidence	0%
ISP	DigitalOcean, LLC
Domain	digitalocean.com
Analyst Note	0% confidence does NOT mean safe — attackers regularly rotate IPs to avoid detection. The presence of 10 reports is still a red flag for a supposed bank mail server.

3.4 Screenshot Evidence — IP Analysis

Figure 1 — WHOIS/ARIN lookup showing IP 137.184.34.4 registered to DigitalOcean cloud hosting

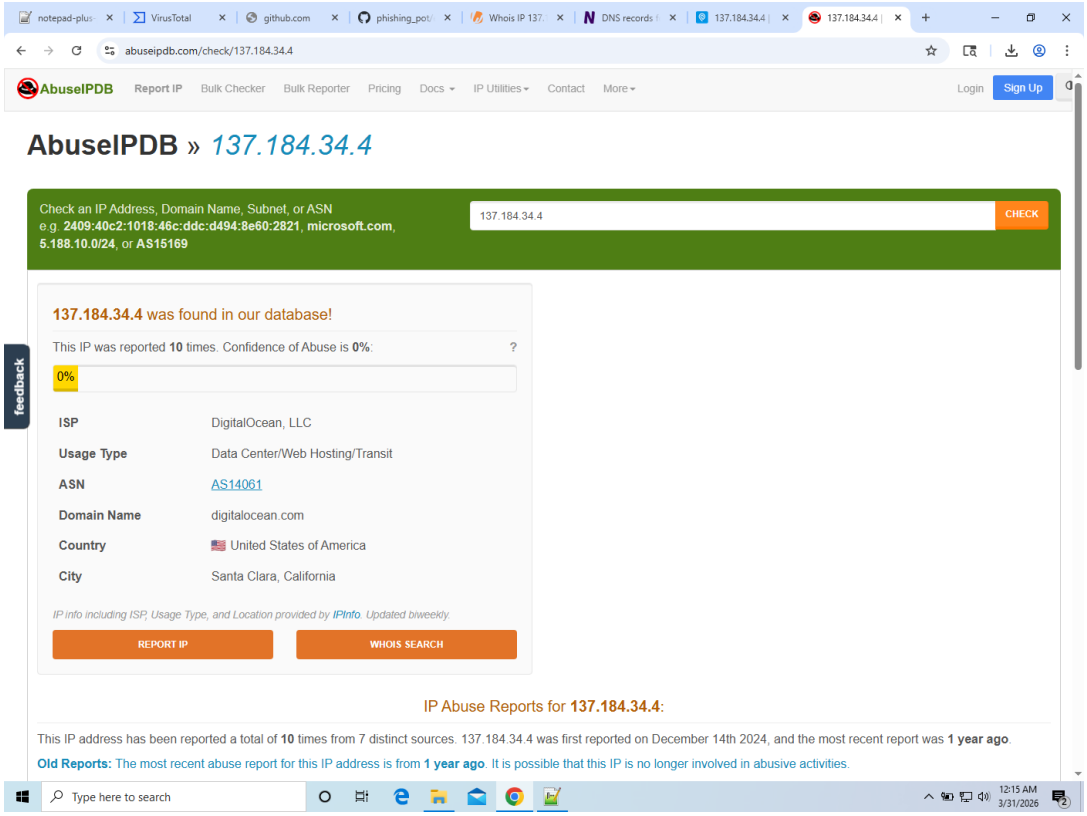

Figure 2 — AbuseIPDB result showing 10 reports for 137.184.34.4 from DigitalOcean

4. Email Content Analysis

4.1 Base64 Encoding — Obfuscation Technique

The email body was Base64 encoded — a common technique used by attackers to bypass email filters and antivirus scanners that analyse plain text content.

Base64 encoding converts binary/text data into ASCII characters. When decoded using CyberChef, the actual HTML phishing content is revealed. Legitimate bank emails do not need to encode their content in Base64.

4.2 Decoded Email Content — English Translation

Translated content: "You have Livelo points with your Banco do Bradesco card available for redemption that expire TODAY. Avoid losing these points by redeeming your Visa Infinite points right now. Exchange your points for airline miles. Discounts of up to 35% on your card statement. 92,990 points accumulated expire TODAY — Redeem Now before they expire!"

4.3 Finding #4 — Social Engineering Tactics

Tactic Used	Description
Urgency creation	"Expiring TODAY" — forces victim to act immediately without thinking
Fear of loss	"Avoid losing these points" — psychological pressure
Large reward bait	"92,990 points" — makes the reward seem very valuable
Brand impersonation	Uses Bradesco and Livelo brand names, logos and style
Call to action	Prominent "Redeem Now" button linking to phishing domain
Legitimacy signals	References to Visa Infinite, airline miles — adds false credibility

4.5 Screenshot Evidence — Content Analysis

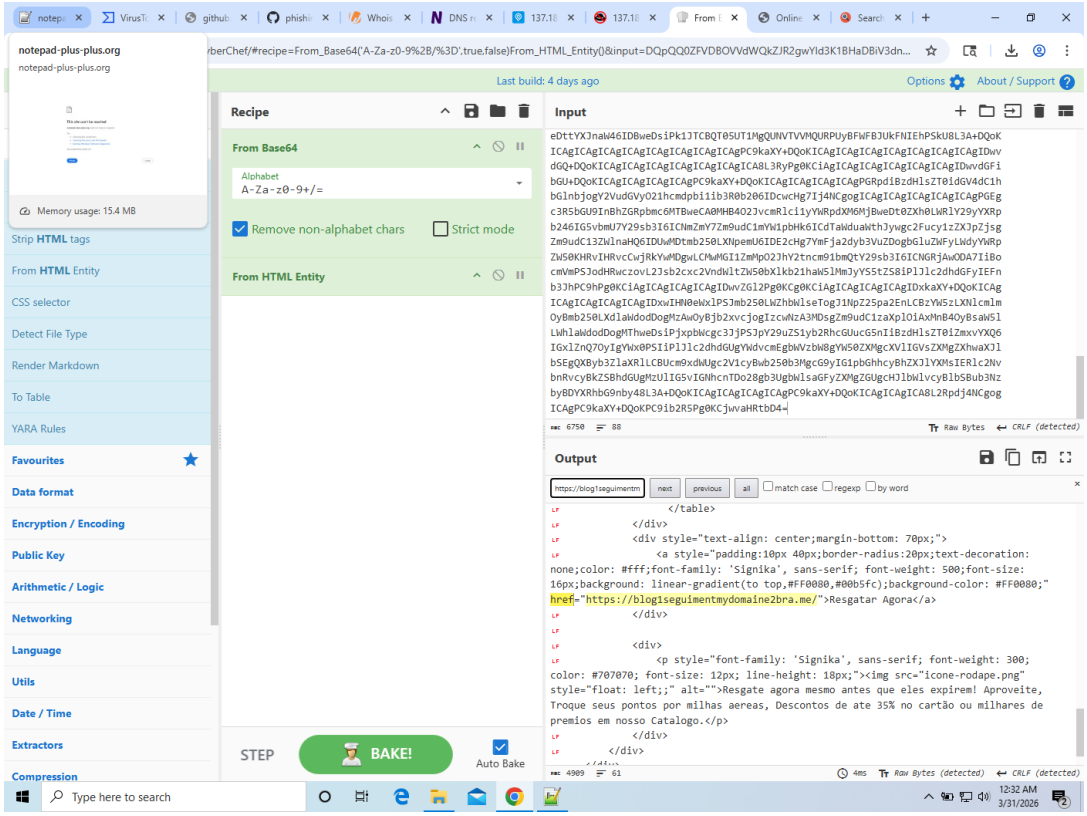

Figure 3 — CyberChef Base64 decode showing the hidden HTML phishing content

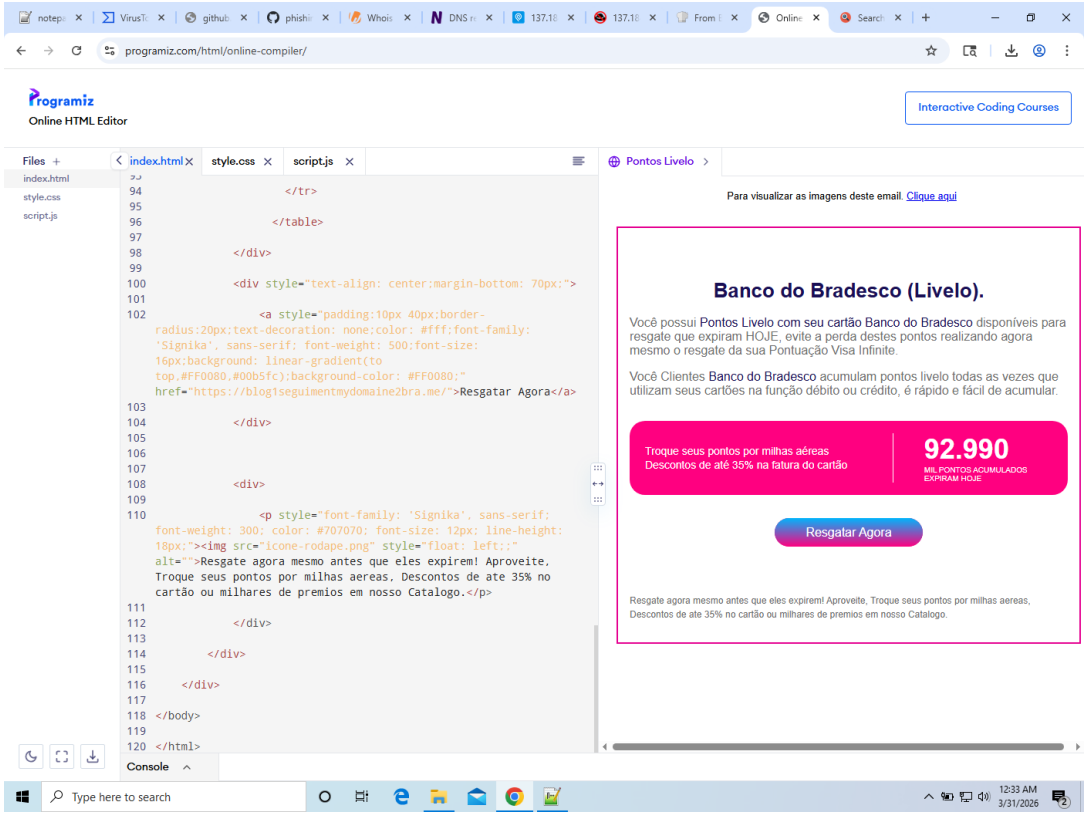

Figure 4 — Decoded HTML output revealing the full phishing email body with malicious links

5. Malicious URL Analysis

5.1 Phishing URL

Malicious URL identified: https://blog1seguimentmydomaine2bra.me/lander

Field	Analysis
Full URL	https://blog1seguimentmydomaine2bra.me/lander
Domain	blog1seguimentmydomaine2bra.me
Path	/lander — classic phishing landing page path
TLD	.me — not a Brazilian domain (.com.br) — red flag
Domain pattern	Contains 'bra' to suggest Brazil/Bradesco but is completely fake
Domain registration	Recently registered — typical of phishing infrastructure
Verdict	MALICIOUS — credential harvesting landing page

5.2 Finding #5 — Domain Mismatch

Comparison	Value
Claimed brand	Banco Bradesco (legitimate Brazilian bank)
Legitimate domain	bradesco.com.br — official bank website
Phishing domain	blog1seguimentmydomaine2bra.me — completely unrelated
Key difference	No legitimate bank link to a .me domain with random words
Victim deception	URL deliberately contains 'bra' to confuse victims who do not look carefully

5.3 Recommended URL Investigation Tools

For future analysis, these tools should be used to fully investigate phishing URLs:

- URLScan.io — scans URL and shows screenshot of landing page and DNS records
- VirusTotal — checks URL against 80+ security vendor databases
- Google Safe Browsing — checks if URL is in Google's threat database
- Any.run — interactive malware sandbox — execute URL safely

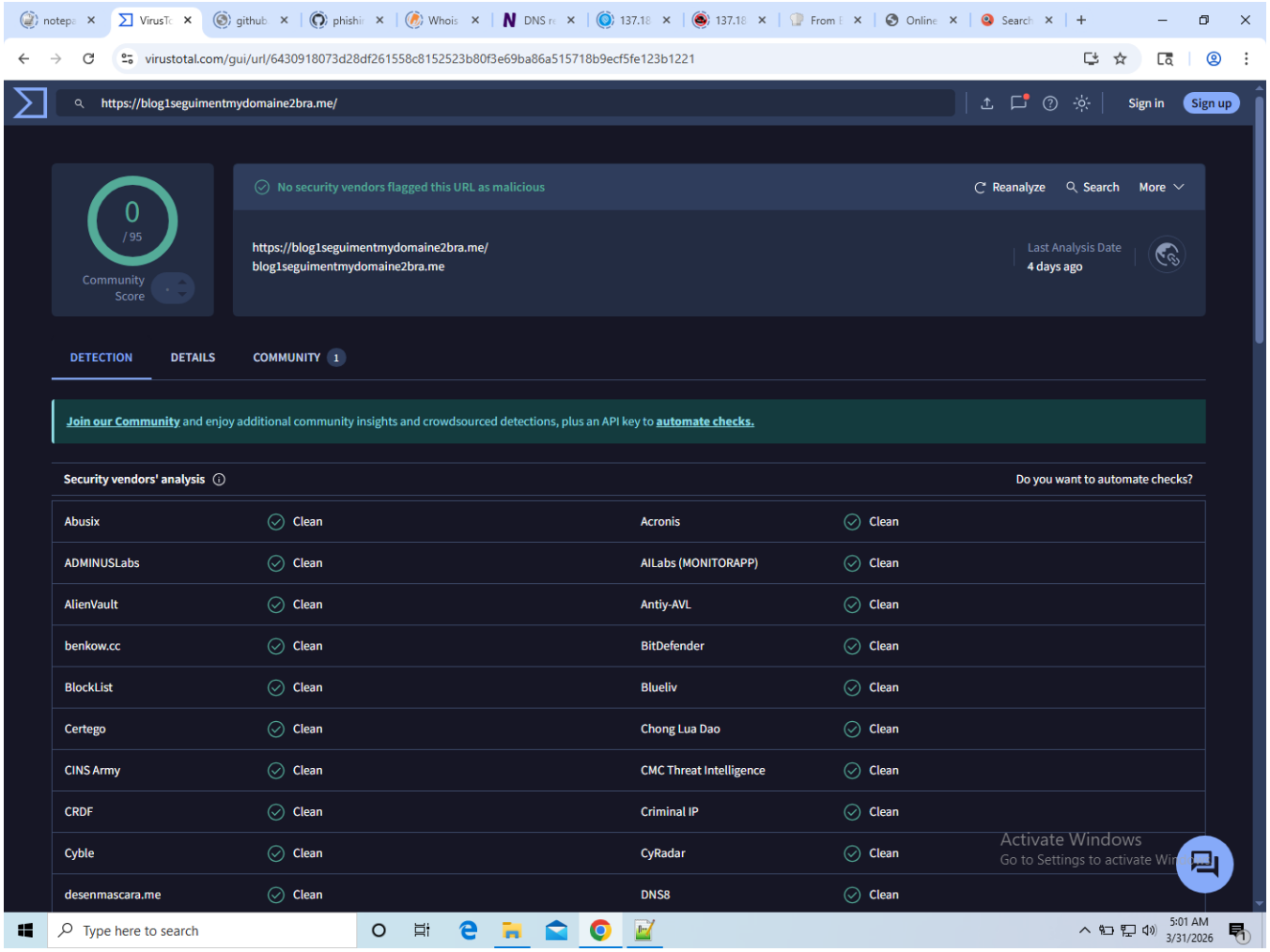

Figure 5 — VirusTotal Result

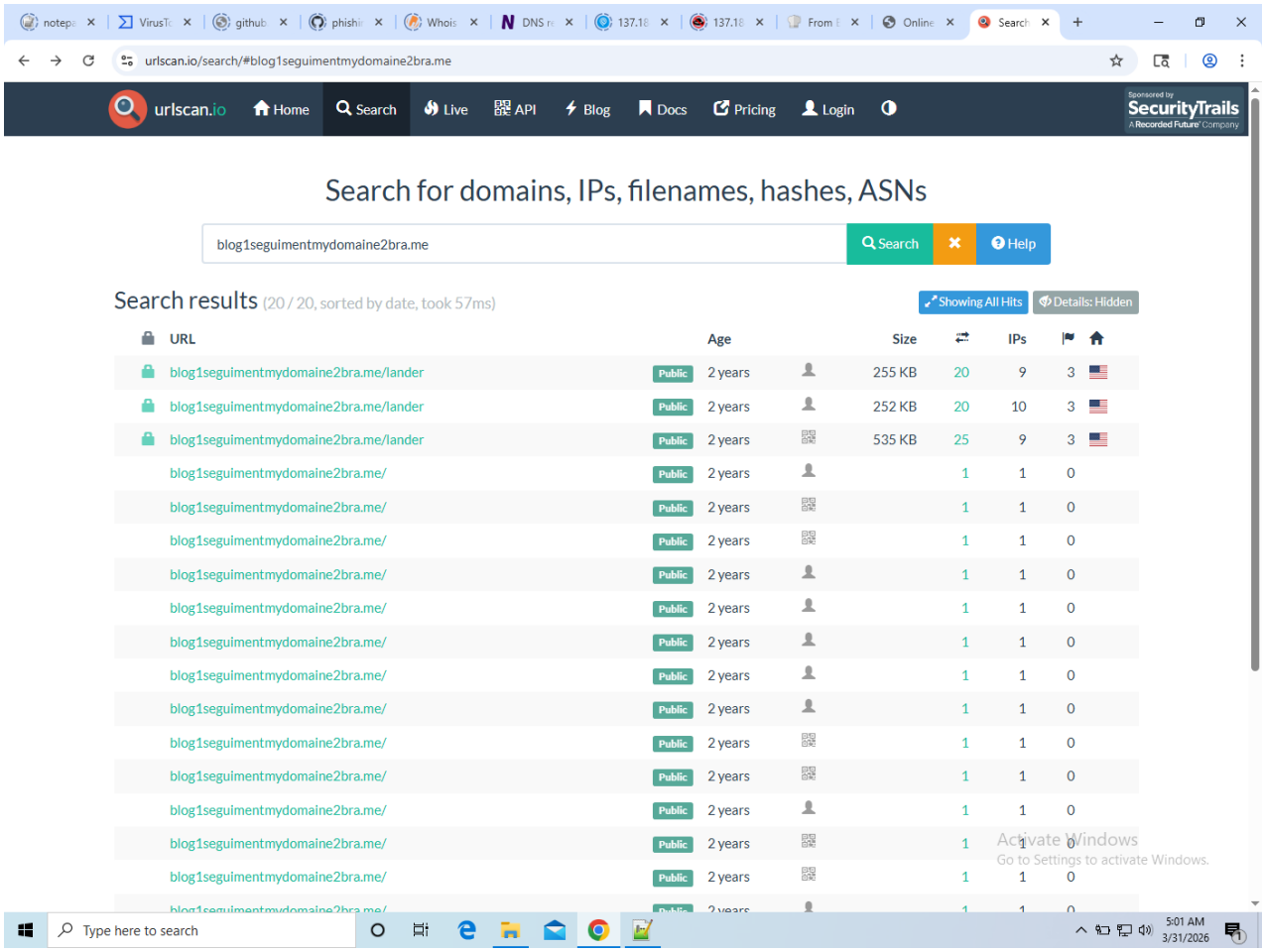

Figure 6 — Urlscan.io Result

6. MITRE ATT&CK Framework Mapping

All identified attack techniques have been mapped to the MITRE ATT&CK framework — the industry standard for classifying cyber threats.

Technique ID	Technique Name	Description	Evidence
T1566.001	Spearphishing Link	Email with malicious link to credential harvester	Phishing URL found in body

T1598.003	Phishing for Information	Gathering credentials via fake bank portal	Landing page /lander
T1204.001	User Execution: Malicious Link	Victim must click link to activate attack	Redeem Now button
T1056.003	Web Portal Capture	Fake web form to steal credentials	blog1seguiment...me
T1036.005	Masquerading: Match Legit Name	Impersonating Bradesco bank brand	Sender display name
T1027	Obfuscated Files	Base64 encoding to evade email filters	Content-Transfer-Encoding

7. Indicators of Compromise (IOCs)

The following indicators can be used to detect and block this phishing campaign across security tools, firewalls, and email gateways.

IOC Type	Value	Tool Used to Identify
IP Address	137.184.34.4	WHOIS ARIN + AbuseIPDB
Hosting Provider	DigitalOcean ASN AS14061	WHOIS ARIN
Sender Domain	atendimento.com.br	Email header analysis
Return-Path	root@ubuntu-s-1vcpu...	Email header analysis
Phishing Domain	blog1seguimentmydomaine2bra.me	Email body URL extraction
Phishing URL Path	/lander	Email body URL extraction
Email Subject	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão...	Email subject line
Encoding Method	Base64 encoded HTML	CyberChef decode
Reward Lure	92,990 Livelo points expiring today	Email content analysis

8. Recommendations

8.1 Immediate Blocking Actions

Block these immediately across all security controls to prevent further phishing attempts from this campaign.

1. Block IP address 137.184.34.4 at the firewall and email gateway
2. Block sender domain atendimento.com.br at the email security gateway
3. Block domain blog1seguimentmydomaine2bra.me at DNS / web proxy
4. Submit IP 137.184.34.4 to AbuseIPDB with full report details
5. Report phishing URL to Google Safe Browsing and VirusTotal

9. Final Conclusion

Based on comprehensive analysis across all investigation dimensions, this email is conclusively confirmed as a high-risk phishing attack.

Finding	Verdict
Sender domain spoofing	CONFIRMED — atendimento.com.br ≠ bradesco.com.br
SPF authentication	FAILED — server not authorised to send
DKIM authentication	FAILED — email not signed
DMARC authentication	FAILED — policy violated
IP origin	SUSPICIOUS — DigitalOcean cloud VPS
Phishing domain	CONFIRMED — blog1seguiment...me is malicious
Social engineering	CONFIRMED — urgency, reward bait, impersonation
Content obfuscation	CONFIRMED — Base64 encoded to evade filters
MITRE techniques	6 techniques mapped — T1566, T1598, T1204, T1056, T1036, T1027

FINAL RISK LEVEL: HIGH — CONFIRMED PHISHING ATTACK

9.1 Resume Summary for Portfolio

Phishing Email Analysis Project | 2026

- Investigated real phishing email from PhishingPot dataset impersonating Banco Bradesco (Brazil)
- Identified sender domain spoofing and complete SPF, DKIM, DMARC authentication failure
- Decoded Base64 encoded HTML payload using CyberChef to reveal hidden phishing content
- Performed WHOIS, AbuseIPDB analysis on attacker IP 137.184.34.4 (DigitalOcean cloud VPS)
- Identified malicious phishing URL blog1seguimentmydomaine2bra.me/lander
- Mapped 6 attack techniques to MITRE ATT&CK: T1566, T1598, T1204, T1056, T1036, T1027
- Documented 9 IOCs with blocking recommendations for firewall, DNS and email gateway

End of Report — Phishing Investigation PHISH-2026-001 —Amit Pal